MACROECONOMIC SCENARIO PAPER | UNITED KINGDOM

The AI-Led Reindustrialisation of Britain, 2027

GDP growth, money multipliers, supply-chain automation and the potential dividends of strategic economic autonomy

CENTRAL CONDITIONAL FORECAST
1.7%-2.1% real GDP growth in 2027, with a central point of about 1.9%, if technology investment is deployed into real UK production, AI adoption deepens and domestic supplier capacity expands. This is a scenario estimate, not an official forecast.

Research question. Can a faster diffusion of AI, a £100bn assumed technology-start-up financing envelope, more automated supply chains and selective import substitution lift UK growth above conventional forecasts in 2027?

Core thesis. Britain can increase the domestic output generated by each pound of investment by combining productivity growth with a deliberate expansion of competitive domestic supply. The argument is for selective resilience, not blanket autarky: foreign competition, imported intermediate goods and access to export markets remain important to productivity.

Prepared 13 October 2026
Evidence base: ONS, OBR, IMF and UK Government research available by the publication date. All scenario calculations are transparent author assumptions and should not be read as official UK forecasts.

Executive summary

The UK enters 2027 with a plausible growth opportunity but no guarantee of an AI-driven boom. The IMF forecast released on 13 October 2026 puts growth at 1.2% in 2027, while the OBR's March 2026 forecast expected average annual growth of 1.6% over 2027-2030. Recent ONS estimates showed GDP rising 0.4% over the three months to July 2026 and 1.3% year-on-year, with services leading the expansion and production and construction weaker over the three-month period. [1-3]

This paper proposes a conditional, higher-growth path of 1.7%-2.1% in 2027. Its logic is not that technology finance automatically becomes GDP. Rather, financing must be converted into UK-based wages, software, research, construction, equipment, commercial services and, ultimately, productive capacity. The £100bn technology-start-up figure supplied for this scenario is treated as a hypothetical gross financing envelope; it has not been verified as realised UK start-up investment in 2026.

A transparent open-economy Keynesian calculation gives a baseline expenditure multiplier of around 1.31 under illustrative assumptions. If the marginal propensity to import falls from 25% to 18% as some supply chains become more domestically supplied, the model multiplier rises to around 1.44, an increase of approximately 10%. That is an accounting illustration, not evidence that a broad import-substitution programme would automatically raise welfare or GDP.

Under the illustrative £100bn financing scenario, assumed UK spending of 55% and additionality of 50% create an effective initial spending impulse of £27.5bn. At the two multipliers, the model produces an implementation-period demand effect of about £36bn-£40bn. If 35% of that effect occurs in 2027, the indicative 2027 impulse is about £13bn-£14bn, or roughly 0.4%-0.5% of an economy scaled at £3tn. It is not a precise national-accounts forecast and must not be added mechanically to productivity estimates: the channels overlap.

The Brexit-related proposition is therefore conditional. Policy autonomy could be used to accelerate planning, compute and energy infrastructure, reform procurement, support experimental AI regulation and improve capital formation. But autonomy is an option, not itself a productivity gain. The OBR continues to assume the current UK-EU trading relationship reduces long-run productivity by 4% and trade volumes by around 15% relative to its remain-in-the-EU counterfactual. [6]

Main findings

- 2027 scenario: 1.7%-2.1% real GDP growth if AI deployment and domestic investment meaningfully outperform current baseline assumptions.
- Multiplier: reducing import leakage from 25% to 18% raises the illustrative multiplier from 1.31 to 1.44, holding other assumptions constant.
- £100bn premise: finance is not GDP. The calculation discounts for imports, non-additional investment, financial transactions and timing.
- Strategic autonomy: target sectors where domestic scale, productivity or security can justify localisation; preserve trade and access to competitive imports.
- Brexit dividend: possible only through measurable reforms and investment responses that more than offset continuing trade frictions.

1. Starting point: the 2027 growth baseline

A forecast for 2027 has to start from the latest baseline rather than assume that prospective technology gains have already been realised. As of 13 October 2026, the IMF projects 1.2% UK real GDP growth in 2027, citing higher energy prices and long-term interest rates as restraints on consumption and investment. The OBR's March 2026 forecast has growth averaging 1.6% a year between 2027 and 2030, while its assumptions also anticipate productivity gradually strengthening. These numbers differ in vintage and horizon, so they are reference points rather than directly interchangeable estimates. [1-2]

The ONS estimates GDP grew 0.4% over the three months to July 2026 and 1.3% compared with the same three-month period a year earlier. Services output grew 0.6% over the three-month period; production and construction each fell 0.5%. Information and communication and professional, scientific and technical activities were among the stronger service subsectors. The data suggest that a technology-led growth strategy has an existing base in knowledge-intensive services, but also needs to bridge the gap between service-sector dynamism and domestic industrial capacity. [3]

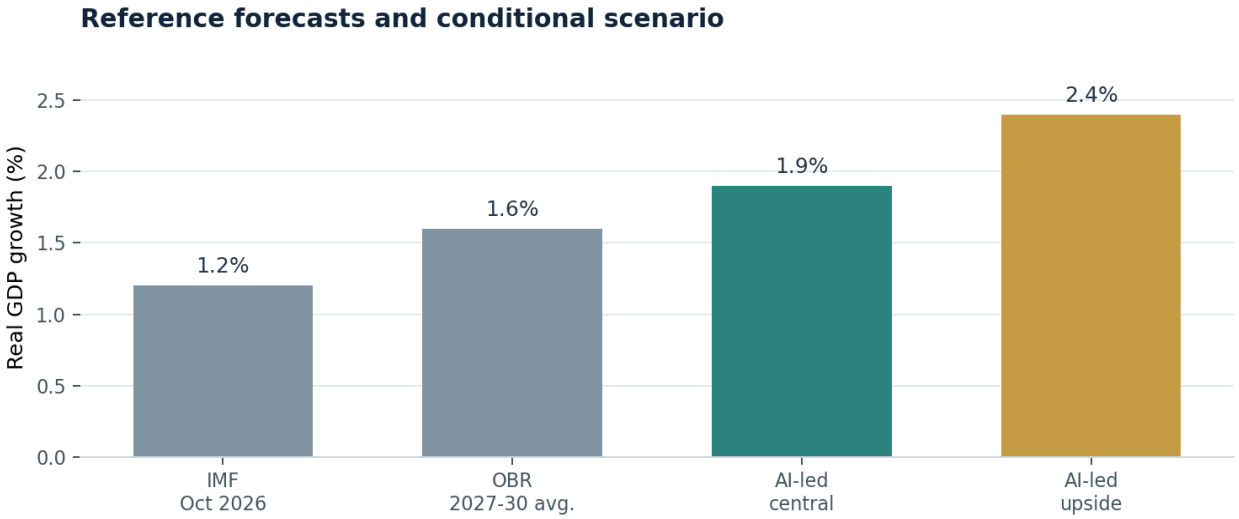


Figure 1. Forecast reference points and this paper's conditional scenarios. The OBR bar refers to average annual growth over 2027-2030, not a separate point forecast for 2027. The AI-led central and upside bars are scenario assumptions, not published official forecasts.

Forecast range proposed by this paper

Scenario	2027 real GDP growth	Conditions
Downside	0.5%-1.2%	High energy and interest costs; weak demand; start-up capital delayed or directed to imported inputs and financial assets; AI adoption remains shallow.
Reference range	1.2%-1.6%	IMF 2027 estimate and OBR medium-term trajectory used as benchmarks, subject to different forecast vintages.
AI-led central scenario	1.7%-2.1%	Technology capex is deployed domestically; automation diffuses beyond large firms; productivity and demand gains partly reinforce one another.
Upside scenario	2.2%-2.6%	Rapid diffusion, strong investment additionality, lower energy costs, efficient planning and power-grid delivery, and exportable UK tech services.

2. AI, automation and the productive capacity of the UK

AI can lift GDP through several distinct channels: more output per hour; new products and services; lower errors and transaction costs; better matching of labour and capital; and accelerated research and design. Supply-chain automation adds a more operational route to productivity: demand forecasting, warehouse robotics, route planning, predictive maintenance, invoice matching and automated procurement can reduce idle inventory, stock-outs, spoilage, delivery time and working-capital needs.

The adoption data support both the opportunity and the caution. The ONS reported in July 2026 that around 35% of UK businesses with 10 or more employees said they used at least one AI technology, up from around 12% in late 2023. However, adoption remained relatively shallow: the average number of technologies used by each adopting firm rose only from about 1.4 to 1.6. Adoption was also uneven by size, from 28% among firms with 0-9 employees to 49% among firms with 250 or more employees. [4]

The OBR's July 2026 long-term fiscal analysis gives a central estimate that AI could increase annual UK productivity growth by around 0.2 percentage points over the next decade, while considering a wide range of outcomes, including under 0.1 points and up to 0.8 points a year. The OBR explicitly regards the effect as highly uncertain. The number is a useful scale check: it does not justify assuming a dramatic productivity step-change in a single year. [5]

A plausible 2027 route is therefore a modest aggregate productivity gain from broad diffusion, with larger benefits building across several years. The strongest early adopters could be professional and business services, software, finance, logistics, retail distribution, advanced manufacturing, food processing and energy systems. The macroeconomic effect depends on management redesign as much as software purchase: firms need data quality, training, secure digital infrastructure and authority to change processes.

From efficiency to new output

Automation can release worker time. GDP rises most durably when that time is redeployed into more output, better service, new products, shorter lead times or additional sales, rather than simply removed from payrolls while capacity sits idle. The transition can involve displacement in specific tasks and occupations; short-term income losses would restrain consumption unless new roles, lower prices, profit-sharing or retraining spread the gains. This paper assumes a net positive output effect, not that all labour displacement is frictionless.

For supply chains, the key metric should be total landed cost and delivery reliability, not domestic content alone. A UK supplier that delivers reliable, competitively priced components can reduce inventory buffers and exposure to disruption. A higher-cost domestic substitute that depends on imported machinery and expensive inputs may deliver little net resilience. A digitally integrated supply chain combines global sourcing where it is efficient with domestic capacity where lead time, security, energy, quality or total cost favour it.

The sector mix matters. New technology firms create direct value through software, engineering and services, but their larger macroeconomic contribution can be indirect: diffusing AI into the much broader stock of existing British firms. The model should therefore measure adoption and process redesign across the economy, rather than treat start-up valuations as the main measure of success.

3. The £100bn technology-start-up assumption

This paper interprets the proposed £100bn in 2026 as a hypothetical gross financing envelope for new technology businesses and related investment. It is not independently verified as realised UK start-up funding. A report published on 13 October 2026 cited $23.3bn of year-to-date funding drawn by UK start-ups in its dataset. That measure is not comprehensive and may omit corporate R&D;, internal capex, private-market transactions and other technology spending; nevertheless, it is not comparable to £100bn without a much wider definition. For that reason, £100bn is used only as a scenario input. [7]

There are four accounting distinctions. First, buying equity in an existing business or an existing asset is a financial transaction, not by itself new GDP. Second, a company raising capital may hold cash rather than spend it during the year. Third, new spending on imported servers, accelerators or equipment generates less immediate UK value added than equivalent spending on UK-produced equipment and services, although imports may still enable later productivity. Fourth, some financing replaces other investment that would have happened anyway. Only the additional production and value added resulting from the finance should be counted as incremental domestic output.

Illustrative conversion from financing to a demand impulse

Step	Assumption	Calculation	Result
Gross technology-start-up financing	Scenario input	Given, not verified as an observed total	**£100.0bn**
Domestic spend over implementation period	55%	£100.0bn × 0.55	**£55.0bn**
Additionality of spend	50%	£55.0bn × 0.50	**£27.5bn**
Baseline multiplier	k = 1.31	£27.5bn × 1.31	**£36.0bn**
Lower-import multiplier	k = 1.44	£27.5bn × 1.44	**£39.6bn**

The 55% domestic-spend assumption represents UK payroll, UK suppliers, research, construction and domestic services over the implementation window; the rest may be imported spending, retained cash, overseas activity or non-GDP financial transactions. The 50% additionality assumption discounts the share of spending that replaces other finance or would have occurred without the headline funding. Both are explicit modelling assumptions, not measured parameters.

To illustrate timing, assume 35% of the cumulative implementation-period demand effect lands in 2027. The baseline case then implies about £12.6bn of demand/output effect in 2027; the lower-import case implies about £13.9bn. Relative to a rough £3tn economy used only as a scale denominator, that is around 0.42%-0.46%. The estimated improvement caused by the smaller import leakage alone is about £1.3bn in 2027, or roughly 0.04 percentage points of that scale denominator.

THE MODEL IS NOT A GDP PROMISE

These numbers are sensitive to the actual amount raised, deployment dates, imported content, additionality, spare capacity, monetary policy and price responses. A gross financing announcement cannot be translated one-for-one into GDP. The calculation illustrates an order of magnitude, and its demand component overlaps with the later productivity gains from the same investment.

4. Money multipliers and domestic circulation

The relevant calculation here is the Keynesian expenditure multiplier, not the older textbook money-supply multiplier that links the monetary base to broad money under simplifying reserve assumptions. The expenditure multiplier estimates how an initial increment in autonomous demand can induce successive rounds of income and spending. For an open economy, imports leak part of each spending round abroad.

An illustrative open-economy formula is:

$$k = 1 / [1 - c(1 - t) + m]$$

where c is the marginal propensity to consume out of disposable income; t is the effective marginal tax rate; and m is the marginal propensity to import from each additional pound of spending. This simplified formula holds these behavioural parameters constant and omits endogenous interest-rate, exchange-rate, capacity and inflation feedbacks.

Case A: current import leakage

Assume c = 0.65, t = 0.25 and m = 0.25.

$$k = 1 / [1 - (0.65 \times 0.75) + 0.25] = 1 / 0.7625 = 1.31$$

Case B: stronger domestic supplier capacity

Keep consumption and tax assumptions unchanged, but assume strategic sourcing and competitive domestic production reduce the marginal propensity to import from 0.25 to 0.18.

$$k = 1 / [1 - (0.65 \times 0.75) + 0.18] = 1 / 0.6925 = 1.44$$

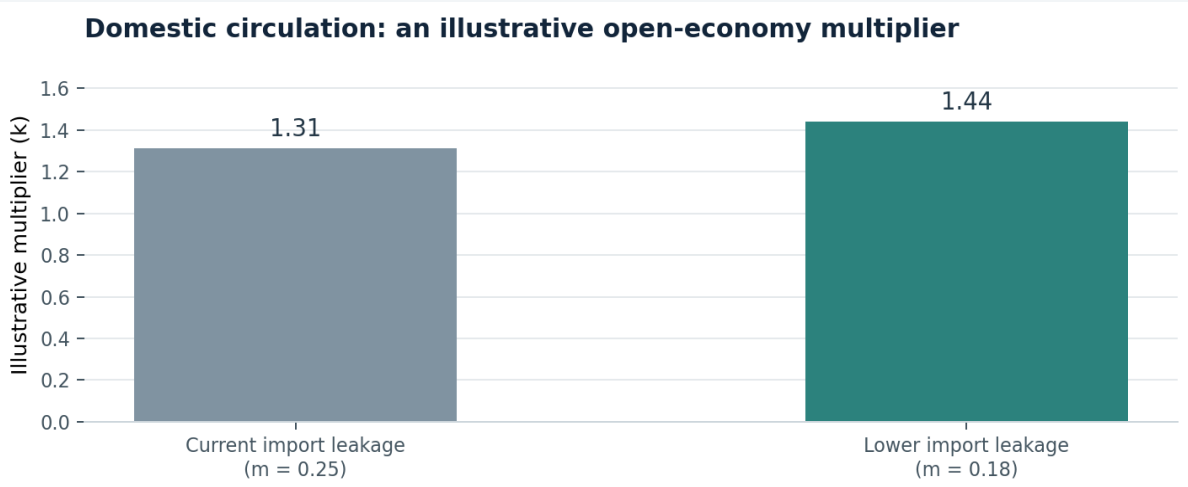


Figure 2. The model multiplier rises by approximately 10% when m falls from 0.25 to 0.18, all else equal. This is a parameter sensitivity calculation rather than an estimate of the realised UK multiplier.

Interpretation

For the same £27.5bn effective initial impulse, the simplified model implies £36.0bn of cumulative activity at k = 1.31 and £39.6bn at k = 1.44. The difference is about £3.6bn over the full modelled implementation period. The result shows why domestic value added can matter for the local propagation of demand; it does not mean any reduction in imports is automatically beneficial.

5. Brexit, strategic autonomy and selective autarky

The economic argument for using post-Brexit autonomy is strongest when it is specific and measurable. The UK can choose how to organise planning approval, procurement, AI experimentation, research support, technical standards, competition policy, energy infrastructure and some elements of taxation and capital formation. Such choices may allow faster iteration or better alignment between regulation and emerging technologies. But any gain must come from better outcomes: faster investment, lower unit costs, new products, more competition or higher productivity. Regulatory divergence alone does not produce output.

Selective domestic sourcing must also pass a cost test. A higher domestic share can mean higher prices, less variety, reduced competitive pressure, lost scale economies or more expensive intermediate inputs. Imports are subtracted from expenditure GDP because imported production is not UK output, but imported inputs can make UK production more productive and enable exports. The objective should be competitive domestic supply where it lowers total cost or reduces material risk, not maximal self-sufficiency at any price.

The official counterfactual is a necessary constraint on any claim that Brexit benefits are now accelerating. The OBR's assumptions for the current UK-EU Trade and Cooperation Agreement put long-run productivity around 4% lower than if the UK had remained in the EU, principally because of additional non-tariff barriers and weaker exploitation of comparative advantage. The OBR also assumes long-run UK export and import volumes each around 15% lower than in that counterfactual. It considers the impacts of new third-country trade agreements gradual and modest; for example, its cited assessments put the UK-Japan and UK-Australia deals at around 0.1% of GDP over 15 years relative to their respective counterfactuals. [6]

Those are modelling assumptions against a counterfactual, not a measure of every sector-specific effect, and the OBR acknowledges that isolating Brexit from the pandemic, energy shock and other developments is difficult. Nevertheless, the paper cannot responsibly treat a net Brexit dividend as already demonstrated. Instead, it models a possible policy-autonomy dividend only if concrete reforms increase productivity or crowd in investment enough to offset trade frictions.

From blanket autarky to strategic resilience

The most credible approach is selective localisation in domains where a domestic cluster can reach adequate scale, quality and price, and where supply security has a demonstrable value. Examples include:

Domain	Opportunity	Test for investment
AI compute and data infrastructure	Domestic compute capacity, model services, secure cloud and energy-efficient data centres.	Power access, utilisation, chip/equipment supply, cooling and customer demand.
Food and controlled-environment agriculture	Greenhouses, hydroponics and automation for high-value crops where UK climate and energy economics allow.	Full-year yield, energy intensity, labour savings, water use and import parity price.
Advanced manufacturing	Robotics, precision components, machine tools, sensors and industrial software.	Total landed cost, skilled workforce, export potential and supplier cluster density.
Energy and critical inputs	Grid equipment, storage, power electronics, selected fuels and industrial materials.	System cost, reliability, lifecycle cost and exposure to external disruption.
Logistics, health and public services	Automated warehousing, inventory systems, scheduling and digital administration.	Measurable reductions in delay, waste and cost without lower service quality.

6. How the forecast is constructed

Taking 1.4% as a midpoint of the current 1.2%-1.6% reference range, the conditional central forecast of 1.7%-2.1% implies a net uplift of roughly 0.3-0.7 percentage points. The channels below are judgemental ranges designed to show where the uplift might come from. They are not independently estimated effects, and their gross sums would double-count interactions unless an overlap adjustment is made.

Channel	Illustrative contribution to 2027 growth	Mechanism
AI diffusion and workflow redesign	+0.10 to +0.20 pp	More output per hour and improved service delivery, conditional on adoption moving beyond pilots.
Supply-chain automation	+0.10 to +0.20 pp	Lower inventory, downtime, spoilage, fulfilment cost and working-capital requirements.
Technology investment and multiplier	+0.20 to +0.40 pp	Incremental domestic spend during deployment; sensitive to the assumed £100bn financing envelope.
Selective import substitution	+0.05 to +0.10 pp	More local value added in competitive strategic sectors and lower exposure to bottlenecks.
Policy autonomy dividend	0.00 to +0.10 pp	Only if targeted reforms bring forward profitable private investment and measured efficiency gains.
Overlap / implementation adjustment	-0.15 to -0.30 pp	Prevents counting the same investment as both current demand and a productivity effect; accounts for delays.
Net scenario uplift	**+0.30 to +0.70 pp**	Applied to a 1.4% midpoint reference baseline gives around 1.7%-2.1%.

Policy priorities for the 2027 window

- Accelerate power and compute. Prioritise grid connections, dependable low-carbon and firm power, data-centre planning and transparent access rules. AI investment without affordable power may shift abroad or displace other productive users.
- Fund deployment, not just invention. Help existing small and medium-sized firms adopt proven automation through demonstrator projects, shared technical services, workforce training and measurable payback criteria.
- Measure full supply-chain cost. Use landed cost, resilience, delivery time and domestic value added together, rather than crude import-replacement targets.
- Keep competition and trade open. Regulatory autonomy should enable rapid experimentation while maintaining high standards and reducing avoidable frictions with major markets.
- Mobilise growth capital. Encourage patient scale-up capital and commercially disciplined R&D;, while monitoring whether financing results in additional UK-based assets, output, exports and skilled work.

Risks and falsification tests

The central scenario should be revised down if UK start-up funding is far below the assumed envelope, capital spending is mostly imported or held as cash, power and planning bottlenecks delay construction, AI adoption remains confined to low-impact use cases, or inflation and interest rates suppress household and business demand. It should be revised up only if measured labour productivity, output per worker, business investment, business formation and export receipts confirm the acceleration. A rise in technology valuations alone is not enough.

Other risks include imported energy-price shocks, higher long-term yields, weak foreign demand, insufficient skills, cyber incidents, market concentration and an uneven distribution of gains. The IMF specifically flags higher energy prices and long-term interest rates as headwinds for UK growth in 2027. [1]

7. Conclusion: a conditional British growth dividend

A more productive UK economy in 2027 is plausible if artificial intelligence becomes a general business tool rather than a narrow technology-sector phenomenon. The key transmission mechanism is a combination of direct investment demand, wider productivity improvements, automation of supply chains and the creation of new tradable services. A policy framework that makes it easier to build energy and digital infrastructure, deploy capital and scale firms can amplify these effects.

The multiplier calculation captures one part of the thesis. With illustrative parameters, lowering the marginal propensity to import from 25% to 18% raises the simplified open-economy expenditure multiplier from 1.31 to 1.44. But this is a statement about the propagation of an initial spending impulse, not proof that autarky maximises national welfare. Competitive imports remain valuable, and domestic firms must deliver quality and price that justify scaling.

Likewise, Brexit can be treated as a platform for policy experimentation, but a net economic dividend is not automatic. The official OBR counterfactual continues to imply long-run costs from reduced trade intensity under the present arrangement. To make a plausible positive case, Britain would need to turn regulatory and policy autonomy into observable gains in investment, productivity and new exports, while minimising unnecessary barriers to trade.

The paper's central scenario is therefore 1.7%-2.1% real GDP growth in 2027, compared with a reference range of 1.2%-1.6%. This range is deliberately conditional. Its success depends on real deployment of capital, faster productivity diffusion and effective supply-chain investment; it should be judged against realised data rather than the headline size of a funding announcement.

References and evidence base

[1] Reuters (13 October 2026), “Britain needs to keep deficit reduction on track, IMF says.” Reports IMF forecast of 1.2% UK GDP growth in 2027 and risks from energy prices and long-term interest rates. Reuters article.

[2] Office for Budget Responsibility (March 2026), Economic and Fiscal Outlook. Forecasts real GDP growth to average 1.6% a year from 2027 to 2030 in the central forecast. OBR report.

[3] Office for National Statistics (11 September 2026), GDP monthly estimate, UK: July 2026. Three-month GDP growth, sector output and annual comparison. ONS release.

[4] Office for National Statistics (20 July 2026), Artificial intelligence in UK businesses: 2023 to 2026. AI adoption by business size and the depth of adoption. ONS analysis.

[5] Office for Budget Responsibility (July 2026), Fiscal risks and sustainability, section on AI and productivity. The OBR central estimate is around 0.2 percentage points added to annual productivity growth over the next decade, with substantial uncertainty. OBR report.

[6] Office for Budget Responsibility (updated 20 July 2026), Brexit analysis. Current assumptions on long-run productivity and trade relative to the remain-in-the-EU counterfactual. OBR analysis.

[7] The Times (13 October 2026), “UK venture capital firms raise record $6.9bn.” Reports a dataset showing $23.3bn of UK start-up funding year-to-date in 2026; this is not a comprehensive measure of all technology investment and is not directly comparable to the £100bn scenario assumption. The Times report.

[8] Department for Science, Innovation and Technology (updated 13 February 2026), AI Adoption Research. Survey evidence on AI use and self-reported business impacts; self-assessed gains should not be treated as causal aggregate-productivity estimates. UK Government report.

Methodological note. Forecast ranges, domestic-spend shares, additionality, import propensities, multiplier values and timing assumptions are scenario parameters chosen for transparency. They are not official statistical estimates. In particular, £100bn of start-up finance is treated as a user-specified hypothetical, not verified as an observed 2026 total. The expenditure multiplier is a simplified short-run framework and omits price, capacity, monetary-policy, exchange-rate and general-equilibrium responses.

End of paper